Year 12 Maths - Triangle geometry

Similar questions - Sectors and small-angle approximations

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 The diagram of a pendulum swing is modelled by a sector OAB of a circle with centre O and radius 8 cm, where angle $AOB = 70^\circ$.

(a) Find the length of the chord AB. **[2]**

(b) Find the area of the segment bounded by the chord AB and the arc AB. **[3]**

(c) Find the perimeter of this segment. **[3]**

2 In triangle ABC, angle $A = 60^\circ$, $AB = (6+h)$ cm and $AC = (6-h)$ cm.

(a) Show that $BC^2 = 36 + 3h^2$. **[3]**

(b) Hence show that, when h is small, $BC \approx 6 + \lambda h^2$, where λ is a rational number to be determined. **[4]**

(c) Check your approximation by evaluating both sides when $h = 0.1$. **[2]**

Answers

Question 1

(a) $AB = 2 \times 8\sin 35^\circ = 9.18$ cm (or cosine rule: $AB^2 = 128 - 128\cos 70^\circ$)

(b) $\theta = 70^\circ = 1.2217$ rad. Segment $= \frac{1}{2} \times 8^2(\theta - \sin\theta) = 32(1.2217 - 0.9397) = 9.02$ cm^2

(c) Arc $= 8 \times 1.2217 = 9.77$ cm. Perimeter $= 9.77 + 9.18 = 19.0$ cm (to 3 s.f.)

Question 2

(a) $BC^2 = (6+h)^2 + (6-h)^2 - 2(6+h)(6-h)\cos 60^\circ = 72 + 2h^2 - (36 - h^2) = 36 + 3h^2$

(b) $BC = 6\left(1 + \frac{h^2}{12}\right)^{1/2} \approx 6\left(1 + \frac{1}{2} \cdot \frac{h^2}{12}\right) = 6 + \frac{h^2}{4}$. So $\lambda = \frac{1}{4}$

(c) Exact: $\sqrt{36.03} = 6.00250\ldots$; approximation $6 + 0.0025 = 6.0025$. They agree to 4 decimal places